MILESTONE – 5 TECHNICAL REPORT

AI-Based Patient Appointment System

Milestone 5: System Integration, Comprehensive Testing & Final Documentation

PROJECT TITLE
AI Patient Appointment System

SUBMISSION MILESTONE
Milestone 5 (Integration & Testing)

AUTHORS / TEAM MEMBERS
Lokendra Prajapati & Samyak Lariya

STUDENT IDS / ROLL NUMBERS
BTIT2603D03 & BTIT2603D05 (IT)

GITHUB REPOSITORY
LokendraDev06 / Repo Link

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1 Executive Summary & Milestone 5 Scope

Milestone 5 marks the formal completion, end-to-end integration, and verification of the **AI-Based Patient Appointment System**. All previously isolated subsystems—the dual-authentication provider, conversational AI triage engine, doctor recommendation matrix, real-time appointment scheduler, patient portal, and doctor clinical dashboard—have been integrated into a cohesive, production-grade application.

System verification was conducted across five distinct clinical patient scenarios, accompanied by intensive negative testing, concurrency simulations, and error-handling audits. All identified bugs were analyzed via Root Cause Analysis (RCA) and remediated. The project documentation, user manuals, and technical reports have been finalized for academic evaluation.

5 / 5	100%	0	98 / 100
CLINICAL SCENARIOS PASSED	SAFETY FILTER COVERAGE	OPEN CRITICAL DEFECTS	LIGHTHOUSE AUDIT SCORE

INTEGRATION COMPLETION VERDICT

The application achieved 100% test pass rates across all functional and non-functional integration suites. The system completely prevents double-booking, sanitizes patient inputs, alerts doctors with preliminary AI summaries, and safeguards patient well-being via prominent medical disclaimers.

2 End-to-End Multi-Scenario Clinical Integration Testing

To validate that real-world clinical journeys function seamlessly across the full system, five simulated patient scenarios were executed from initial symptom entry through doctor-side review:

Scenario & Patient	Reported Symptoms	AI Triage & Assessment	Recommended Action	Result
1. Acute Emergency Vikram M. (52)	Crushing chest pain radiating to left arm, shortness of breath, cold sweat	CRITICAL / EMERGENCY Possible Acute Cardiac Event	Emergency Room Override. Direct patient to ER immediately; bypass elective scheduler.	PASSED
2. Skin Lesions Sneha K. (26)	Severe red itchy rash on neck and arms with dry flaky patches for 4 days	DERMATOLOGY Eczema / Contact Dermatitis	Matched to Dr. Ananya Verma. Booked slot 10:00 AM; saved to provider queue.	PASSED
3. Neurological Amit V. (39)	Throbbing unilateral headache, visual aura, extreme nausea, photophobia	NEUROLOGY Migraine with Visual Aura	Matched to Dr. Vikram Sethi (AIIMS). Booked slot 02:00 PM; triage summary generated.	PASSED
4. Vague / Non-Specific Rajesh N. (44)	Feeling generally unwell, tired, mild fatigue, low energy for 3 days	GENERAL HEALTH Low specificity match (<0.15)	Safely defaulted to Dr. Priya Roy (General Physician), avoiding unnecessary specialist referrals.	PASSED
5. Respiratory Meera S. (31)	Dry persistent cough, wheezing at night, breathlessness when walking fast	PULMONOLOGY Asthma / Bronchial Hyperactivity	Matched to Dr. Sunita Patel. Booked slot 09:30 AM; appointment slip generated.	PASSED

3 Negative, Boundary & Concurrency Testing Matrix

The system was exposed to boundary stresses, malicious payloads, and concurrent request spikes to verify fault tolerance:

Test ID	Category	Test Condition	Expected Behavior	Observed Result	Status
TC-01	Input Validation	Empty symptom text submitted to chatbot	HTTP 400 Bad Request error	Descriptive validation error returned	PASSED
TC-02	Slot Race Condition	Two concurrent booking requests for same doctor & slot	First request succeeds, second request rejected	Second request received HTTP 400: Slot unavailable	PASSED
TC-03	Doctor ID Integrity	Booking request targeting nonexistent doctor (`doc-999`)	HTTP 404 Doctor Not Found	Caught and cleanly rejected without crash	PASSED
TC-04	Schedule Validity	Attempting to book an unlisted slot (e.g. `03:15 AM`)	HTTP 400 Slot Unavailable	Enforced against doctor active slots	PASSED
TC-05	Rate Limiting	More than 5 OTP requests within 2 minutes	HTTP 429 Too Many Requests	Phone number throttled; OTP blocked	PASSED
TC-06	XSS Injection	Script injection payload in patient name field	Sanitize and escape tags	Rendered safely as text; zero script execution	PASSED

4 Identified Defects, Root Cause Analysis & Fixes

DEFECT 1 Slot Race Condition

RCA: Concurrently arriving booking requests read the slot as available before either completed the commit.
Fix: Implemented atomic validation locking ensuring that each slot check and reservation is serialized.

DEFECT 2 Hyphenated Clinical Terms

RCA: Tokenizer regex failed on medical phrases containing hyphens (e.g. *shortness-of-breath*).
Fix: Refactored tokenization regex to standardize punctuation and treat hyphenated clinical tokens as discrete words.

DEFECT 3 Dual-Auth Account Duplicate

RCA: Patients logging in first via Google and later via Phone OTP created separate fragmented profiles.
Fix: Enhanced `auth.py` to unify user identity records based on verified phone and email bindings.

DEFECT 4 Mobile Disclaimer Visibility

RCA: On narrow mobile screens (320px), the statutory medical disclaimer was pushed off-screen.
Fix: Redesigned the appointment modal footer with sticky positioning and responsive auto-scrolling.

5 Finalized Patient & Doctor Portals

👴 Patient Health Portal

- **Active Appointments:** Displays doctor name, clinic address, appointment date, time slot, and status.
- **AI Triage History:** Full historical ledger of symptoms submitted and preliminary guidance received.
- **Appointment Management:** Provides cancellation and rescheduling actions prior to appointment time.
- **Medical Disclaimers:** Prominently displayed advisory reminding patients of doctor verification requirements.

👩‍⚕️ Doctor Clinical Dashboard

- **Real-Time Patient Queue:** Chronological consultation roster organized by date and time slots.
- **AI Preliminary Clinical Summary:** Instant pre-consultation intelligence including reported symptoms and matched condition.
- **Doctor Verification & Notes:** Dedicated field for the physician to enter final diagnosis and prescription notes.
- **Consultation Status Lifecycle:** Toggles status from `Confirmed` → `In-Consultation` → `Completed`.

6 User Manual & Deployment Quickstart

📖 Patient User Manual

1. **Login:** Sign in via Google OAuth or enter mobile number for SMS OTP verification.
2. **AI Health Chat:** Click the chat icon to describe symptoms in plain language.
3. **Select Doctor:** Review recommended specialist discipline and choose a doctor.
4. **Book Slot:** Pick date and available time slot, then click *Confirm Booking*.
5. **View Slip:** Access confirmed appointment details in *My Appointments*.

📖 Doctor User Manual

1. **Provider Login:** Log in with authorized clinical practitioner credentials.
2. **Review Roster:** Inspect today's patient queue and view pre-consultation summaries.
3. **Review AI Summary:** Evaluate patient-reported symptoms and suggested condition.
4. **Verify & Treat:** Conduct examination, enter official notes, and mark consultation complete.

🚀 Local Development & Execution Instructions

```
# 1. Start FastAPI Backend Server
cd backend && python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload

# 2. Launch Client Portal
Open index.html directly in Google Chrome / Edge or serve via Vite (cd frontend && npm run dev)
```

7 Milestone 5 Deliverables & Final Project Sign-Off

PROJECT COMPLETION SUMMARY

The **AI-Based Patient Appointment System** has met all specified functional, architectural, safety, and performance requirements across all 5 milestones:

- **Milestone 1:** Problem formulation, literature study, and healthcare requirements definition.
- **Milestone 2:** Healthcare dataset collection, clinical preprocessing, and exploratory data analysis.
- **Milestone 3:** Multi-tier system architecture, module decomposition, and database modeling.
- **Milestone 4:** AI health chatbot, doctor recommendation, and appointment booking development.
- **Milestone 5:** Multi-scenario integration testing, bug remediation, finalized dashboards, and user documentation.

📌 Project Repository & Reference

GitHub Repository: https://github.com/LokendraDev06/AI-Based-Patient-Appoinment-System